

TB Delay

DETAILED ENGLISH USER MANUAL

A stereo feedback-delay instrument with Free/Sync/Tune timing, multiple tap styles, character models, modulation, diffusion, pitch/frequency/granular transforms, ducking, freeze, and routing options.



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Host-visible endpoints documented: 38

1. Product purpose

A stereo feedback-delay instrument with Free/Sync/Tune timing, multiple tap styles, character models, modulation, diffusion, pitch/frequency/granular transforms, ducking, freeze, and routing options.

A conventional delay repeats a fixed copy. TB Delay is designed for repeats that evolve on every cycle, while still covering precise utility echo, slap, dual, ping-pong, ratio, and tempo-locked work. Character, color, modulation, and grain transforms can live inside the feedback loop or after it.

What result it creates

- Clean utility and tempo-synced delay
- Stereo dual, ping-pong, ratio, and multi-tap patterns
- Tape, BBD, and digital character
- Pitch, frequency-shift, reverse, diffusion, and granular evolution

Signal flow

Input -> timing/tap generator -> stereo style and spacing -> feedback loop with character/color/transform routing -> ducking and output -> dry/wet Mix

2. Complete feature guide

Time modes

Free uses Delay Time in milliseconds. Sync uses host tempo and Division. Tune treats delay as a pitch-related comb interval. Time Change selects Crossfade or Glide when timing moves.

Styles and taps

Single, Dual, Ping Pong, and Ratio determine stereo topology. Taps selects 1, 2, 4, or 8 repeats; Spacing, Swing, Offset, Diverge, and Width distribute them.

Feedback and safety

Feedback extends regeneration to self-sustaining territory. Freeze holds the current loop. Change high-feedback and Freeze states carefully because internal energy can accumulate.

Character

Clean, Tape, BBD, and Digital models combine with Drive and Age. Low Cut and High Cut control feedback bandwidth.

Modulation and diffusion

Mod Rate and Mod Depth move delay time. Smear and Size diffuse repeats from discrete echoes toward a cloud.

Spectral transforms

Pitch shifts repeats in semitones, Shift frequency-shifts them in hertz, and Reverse sets reverse probability. These can create ascending, descending, or inharmonic cycles.

Granular system

Grain, Grain Size, Density, Spray, and related transform controls fragment the repeats. Grain Transform Bypass disables that transform block without changing the whole delay.

Transform Route

Loop places transforms inside regeneration so every cycle evolves; Post applies them after the feedback loop for a more stable echo core.

Ducking, Mix, and Output

Duck lowers wet activity around the dry source. Mix sets final dry/wet balance and Output trims the complete result. Stereo Bypass and Color Space Bypass isolate defined subpaths in the deployed build.

Programs

The extensive program bank covers utility delay, vocal pocket, guitars, ping-pong, golden ratio, slap families, doubles, chorus drift, reverse tails, and octave sparks.

3. Quick start

1. Choose Free, Sync, or Tune and set Delay Time or Division.
2. Choose Style, Taps, Spacing, and Width.
3. Set Feedback below self-oscillation first.
4. Choose Character and set Drive, Age, Low Cut, and High Cut.
5. Add modulation or Smear/Size.
6. Choose Loop or Post transform routing, then add Pitch, Shift, Reverse, or Grain.
7. Set Duck, Mix, and Output.
8. Use Freeze only after loop level is controlled.

4. Practical workflows

Vocal dotted eighth

1. Choose Sync and 1/8 D.
2. Use Ping Pong or Dual.
3. High-pass and low-pass the feedback.
4. Add Duck so words remain clear.
5. Keep transforms off or Post for a stable repeat.

Expected result: A wide rhythmic throw supports the vocal without masking each phrase.

Evolving pitch spiral

1. Use moderate Feedback.
2. Set Transform Route to Loop.
3. Add +7 or +12 Pitch and some Smear.
4. Use High Cut and Output to contain accumulating brightness.

Expected result: Every repeat climbs and diffuses into a controlled spiral.

Granular freeze cloud

1. Set a stable loop first.
2. Raise Grain, Density, and Spray.
3. Use Freeze only after the level is safe.
4. Automate Mix and Output for entry and exit.

Expected result: The delay becomes a suspended textured cloud.

5. Automation, gain staging, and session use

- Automate only controls that serve a clear musical transition. Fast movement of frequency, time, pitch, mode, oversampling, or algorithm selectors may intentionally create artifacts or require a host-safe transition.
- Match perceived loudness before judging quality. A louder setting will often appear better even when the processing decision is not better.
- Use the plug-in bypass endpoint for comparison and preserve an unprocessed source or earlier project version.
- Factory programs are starting points. Loading a program changes multiple parameters; save a new DAW preset after adapting it to the source.
- The parameter appendix is captured from the installed VST3 host contract. It lists every host-visible endpoint, its displayed range/options, default, automation status, and identifier.

6. Limits, cautions, and troubleshooting

- Feedback above 100 percent and Freeze can create runaway level.
- Tempo sync requires valid host BPM.
- Pitch, grain, diffusion, and high tap counts increase CPU and may add artifacts intentionally.
- The current host text round-trip for one delay parameter has a minor precision mismatch; audio operation is unaffected.
- No audible change: confirm the plug-in and relevant sub-block are enabled, Mix/Amount is not at a neutral value, and the source contains energy in the processed region.
- Unexpected level: return input, output, send, feedback, and parallel-mix controls to conservative values, then rebuild the setting one block at a time.
- Automation does not match the panel: reload the project, confirm the DAW is addressing the current plug-in version, and check whether a factory program or state recall replaced values.
- Clicks or transitions: avoid sample-instant changes to algorithm, time, pitch, mode, or oversampling selectors unless the sound is intentional.

7. Complete host parameter reference

This appendix contains all 38 parameters exposed by the current installed VST3 to a host. Repeated EQ-band endpoints are intentionally listed rather than collapsed. Discrete options are printed in full when the host contract exposes them.

Parameter	Range / options	Default	Host status	Function
Age VST3 ID 96511	0.00 to 100.00	28.00	Automatable	Host-automatable control for Age. Its full range and default are shown in this table; use it with the related feature section above.
Bypass VST3 ID 1652125811	Off, On	Off	Automatable; Bypass	Turns the complete plug-in processing path off or on through the host-visible bypass endpoint.
Character VST3 ID 1564195625	Clean, Tape, BBD, Digital	Tape	Automatable	Selects the operating algorithm, response shape, or tonal character for the named block.
Color Space Bypass VST3 ID 2141649309	Off, On	Off	Automatable	Selects the operating algorithm, response shape, or tonal character for the named block.
Delay Time VST3 ID 3560141	10.0000000 to 8000.0000000	375.0000305	Automatable	Host-automatable control for Delay Time. Its full range and default are shown in this table; use it with the related feature section above.
Density VST3 ID 1552717032	1 to 8	4	Automatable	Controls granular density, dispersion, or random variation for the named process.
Diverge VST3 ID 1674212956	0.00 to 100.00	0.00	Automatable	Host-automatable control for Diverge. Its full range and default are shown in this table; use it with the related feature section above.
Division VST3 ID 364720301	1/32, 1/32 T, 1/32 D, 1/16, 1/16 T, 1/16 D, 1/8, 1/8 T, 1/8 D, 1/4, 1/4 T, 1/4 D, 1/2, 1/2 T, 1/2 D, 1/1, 1/1 T, 1/1 D	1/8 D	Automatable	Host-automatable control for Division. Its full range and default are shown in this table; use it with the related feature section above.
Drive VST3 ID 95852938	0.00 to 100.00	18.00	Automatable	Controls or enables non-linear density, harmonic color, or peak shaping.
Duck VST3 ID 3094713	0.00 to 100.00	25.00	Automatable	Reduces the effect return while direct input is present so the source remains clear.
Feedback VST3 ID 1955982213	0.00 to 110.00	48.00	Automatable	Returns processed signal to its own input, extending or intensifying the effect.
Freeze VST3 ID 881080983	Off, On	Off	Automatable	Holds or sustains the current internal state until released.
Grain VST3 ID 98615419	0.00 to 100.00	0.00	Automatable	Controls granular density, dispersion, or random variation for the named process.
Grain Size VST3 ID 535555397	10.0000000 to 500.0000000	90.0000076	Automatable	Shapes spatial density, resonant spread, or diffuse body for the named process.
Grain Transform Bypass VST3 ID 1352848479	Off, On	Off	Automatable	Sets how strongly the named process affects the signal.
High Cut VST3 ID 462548517	500.0000000 to 20000.0000000	9000.0000000	Automatable	Reduces high-frequency content or shortens high-frequency decay in the associated path.
Low Cut VST3 ID 356813527	20.0000000 to 2000.0000000	120.0000229	Automatable	Reduces low-frequency content in the associated audio or detector path.
Mix VST3 ID 108124	0.00 to 100.00	30.00	Automatable	Sets the balance between the original signal and the complete processed path.
Mod Depth VST3 ID 2105674566	0.000 to 20.000	0.800	Automatable	Sets how strongly the named process affects the signal.
Mod Rate VST3 ID 1523085309	0.0200000 to 10.0000000	0.2500000	Automatable	Sets the speed of modulation, movement, or repeated change.
Offset VST3 ID 1127703699	-100.00 to 100.00	0.00	Automatable	Host-automatable control for Offset. Its full range and default are shown in this table; use it with the related feature section above.
Output VST3 ID 1141971201	-18.00 to 6.00	0.00	Automatable	Sets or limits the final output level after the plug-in's main processing.
Pitch VST3 ID 106677056	-24.00 to 24.00	0.00	Automatable	Moves pitch, frequency structure, or perceived resonant size for the named process.

Parameter	Range / options	Default	Host status	Function
Program VST3 ID 1886548852	Init, Clean Utility, Vocal Pocket, Wide Dual Guitar, Classic Ping Pong, Golden Ratio, Tape Slap, Dub Throw, Mono Safe Quarter, Clean Dotted Eighth, Studio Slap 80, Vintage Slap 120, Dark Vocal Slap, Bright Guitar Slap, Stereo Double 18, Loose Double 28, Microshift Double, Stereo Chorus Drift, Reverse Tail Slap, Wide ADT, Straight Eighth Grid, Dotted Pulse Train, Triplet Relay, Sixteenth Cascade, Quarter Note Call, Late Swing Taps, Early Swing Taps, Contracting Steps, Expanding Steps, Eight Tap Machine, Narrow Relay, Panoramic Relay, Lazy Right Return, Fast Right Return, Dual Counterstep, Ratio Crosscurrent, Opposed Orbit, Diffuse Ping Pong, Mono Input Expander, Stereo Dub Circle, Fresh Tape Repeat, Warm Reel Echo, Worn Tape, Saturated Tape Loop, Slow Wow Memory, Fast Flutter Reel, Broken Cassette, BBD Haze, BBD Chorus Echo, Dark Bucket Trail, Pristine Digital, Digital Crunch, 12-Bit Repeat, 6-Bit Collapse, Clocked Ping Pong, Telephone Memory, Narrowband Pager, Lo-Fi Stutter, Buffer Skip, Pixel Dust, Comb Bloom, Hollow Notch, Plucked Wire, Glass String, Bass Tube, Damped Resonator, Subtle Flange, Jet Flange, Stereo Comb Split, Digital Bell Bank, Spiral Ascender, Spiral Descender, Mirror Orbit, Metallic Barber Pole, Gentle Lift, Gentle Fall, Opposed Currents, Post Shift Halo, Ping Pong Helix, Grain Shift Dust, Small Room Bloom, Diffuse Chamber, Dark Hallway, Bright Plate Trail, Eight Tap Nebula, Wide Ambient Wash, Mono Fog, Rhythmic Bloom, Deep Space Return, Ducking Atmosphere, Grain Cloud, Reverse Cloud, Freeze Ready Constellation, Scatter Echo, Micrograin Chorus, Tape Granules, Freeze Ready Reverse Swell, Shimmer Twelve, Falling Fifth, Two Octave Spark	Init	Automatable	Selects a factory program. Loading a program recalls a coordinated set of plug-in parameters.
Reverse VST3 ID 1099846370	0.00 to 100.00	0.00	Automatable	Sets whether or how often audio fragments are played in reverse.
Shift VST3 ID 109407362	-2000.00 to 2000.00	0.00	Automatable	Moves pitch, frequency structure, or perceived resonant size for the named process.
Size VST3 ID 3530753	2.0000000 to 120.0000000	32.0000000	Automatable	Shapes spatial density, resonant spread, or diffuse body for the named process.
Smear VST3 ID 109552316	0.00 to 100.00	12.00	Automatable	Shapes spatial density, resonant spread, or diffuse body for the named process.

Parameter	Range / options	Default	Host status	Function
Spacing VST3 ID 135324739	25.00 to 200.00	100.00	Automatable	Host-automatable control for Spacing. Its full range and default are shown in this table; use it with the related feature section above.
Spray VST3 ID 109654189	0.00 to 100.00	15.00	Automatable	Controls granular density, dispersion, or random variation for the named process.
Stereo Bypass VST3 ID 24406799	Off, On	Off	Automatable	Host-automatable control for Stereo Bypass. Its full range and default are shown in this table; use it with the related feature section above.
Style VST3 ID 109780401	Single, Dual, Ping Pong, Ratio	Ping Pong	Automatable	Selects the operating algorithm, response shape, or tonal character for the named block.
Swing VST3 ID 109854462	-50.00 to 50.00	0.00	Automatable	Host-automatable control for Swing. Its full range and default are shown in this table; use it with the related feature section above.
Taps VST3 ID 3552560	1, 2, 4, 8	1	Automatable	Host-automatable control for Taps. Its full range and default are shown in this table; use it with the related feature section above.
Time Change VST3 ID 679287144	Crossfade, Glide	Glide	Automatable	Host-automatable control for Time Change. Its full range and default are shown in this table; use it with the related feature section above.
Time Mode VST3 ID 36460501	Free, Sync, Tune	Sync	Automatable	Selects the operating algorithm, response shape, or tonal character for the named block.
Transform Route VST3 ID 1824145334	Loop, Post	Loop	Automatable	Sets how strongly the named process affects the signal.
Width VST3 ID 113126854	0.00 to 200.00	130.00	Automatable	Sets stereo spread or the lateral distribution of the processed signal.

Documentation basis

Prepared from the current public catalog, current local product brief and implementation notes where available, existing English manuals where available, and a direct scan of the installed VST3 host contract. Internal implementation details that are not user-facing are intentionally excluded; all user-facing features and host-visible controls are documented.